

Conditions on option prices for absence of arbitrage and exact calibration*

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First Version: September 15, 2004

This version: June 22, 2006

Abstract

Under the assumption of absence of arbitrage, European option quotes on a given asset must satisfy well-known inequalities, which have been described in the landmark paper Merton (1973). If we further assume that there is no interest rate volatility and that the underlying asset continuously pays deterministic dividends, cross-maturity inequalities must also be satisfied by the bid and ask option prices.

In this paper, we show that there exists an arbitrage-free model, which is consistent with the option quotes, if these inequalities are satisfied. One implication is that all static arbitrage strategies are linear combinations, with positive weights, of those described here. We also characterize admissible default probabilities for models which are consistent with given option quotes.

JEL classification: C60, G12.

AMS classification codes: 60G44, 60J10.

Key words: options, arbitrage, calibration, default probability.

1 Introduction

Since the 1987 crash, which showed some of the shortcomings of the Black-Scholes model (see Black and Scholes (1973)), numerous models have been proposed to better fit European option quotes. However, those who claim this fitting to be exact are rare.

Among them we should cite local volatility models, described in Derman and Kani (1994) and Dupire (1994) as well as local Lévy models introduced in Carr et al. (2004). But these models assume that European options of all strikes and maturities are traded, which leads in practice to an

*I am grateful to Peter Carr, Bruno Dupire and more generally to the Bloomberg Quantitative Finance Research Team for all their insights. I am also grateful to two anonymous referees for interesting comments. Remaining errors are of course mine.

interpolation of the market option prices. And, to the best of our knowledge, no rigorous algorithm has been described so far to calibrate these models exactly in practice.

One objective of the present paper is to give conditions on quoted European option prices which allow exact calibration by an arbitrage-free model. Notice that these conditions will be necessary for the absence of arbitrage.

This part of the paper has mainly been inspired by Carr and Madan (2005) and we will generalize their results by allowing:

- the underlying asset to pay deterministic dividends continuously,
- deterministic interest rates,
- the options to have any strike or maturity and to be in finite number,
- the options to have different bid and ask prices.

Two recent working papers, Buehler (2004) and Davis and Hobson (2004), have the same goal and a similar treatment. Nevertheless Buehler (2004) does not provide arbitrage strategies which are related to conditions on option prices but instead describes criteria to choose a more realistic model in the subset of calibrated arbitrage-free Markov chain models, as does Cousot (2005). Furthermore, both preprints do not allow different bid and ask prices.

We should also acknowledge the work of Laurent and Leisen (1998) which obtained similar results in spirit by extending the concept of Arrow Debreu securities and by constructing a Markov chain model fitting the market quotes. Nevertheless our treatment is perhaps more rigorous since we attach a great importance to justifying the existence of what would be in their case transition matrices which give rise to martingales. Moreover, like the two working papers cited above, their framework does not encompass the difference between bid and ask prices.

Another objective of the present paper is to characterize the default probabilities which are admissible for the stock among the arbitrage-free models which are calibrated to a given set of option prices. Indeed, with the emergence of credit derivative products such as Credit Default Swaps (CDS), it is more and more desirable for an equity model to be calibrated to both option prices and default probabilities. (See Atlan and Leblanc (2005), Carr and Linetsky (2006) and the references therein for examples of models which attempt to capture both). Nevertheless, to the best of our knowledge, admissible default probabilities have never been characterized.

The structure of the paper is given as follows. Section 2 specifies our assumptions and introduces the definitions of calendar vertical spreads and calendar butterfly spreads. Section 3 describes necessary conditions on option quotes under the assumption of no arbitrage and also discusses briefly why these conditions are not sufficient for the absence of arbitrage. However, Section 4 shows that these conditions are necessary and sufficient for the existence of a calibrated arbitrage-free model. Section 5 describes the set of admissible default probabilities of such a model. Section 6 concludes.

2 Framework of the problem

2.1 Assumptions and notations

We assume that we have at our disposal a finite set of European call option quotes on a given stock¹. Their maturities, indexed in increasing order, are denoted by $(T_i)_{1 \leq i \leq m}$. For a given maturity T_i ,

¹For the sake of simplicity, we consider only call options since a put option can be replicated by holding a call option, bonds and being short a share of stock, what is allowed in our framework as we will see in Section 3.1.

n_i ($n_i \geq 1$) quotes are available. The strikes are denoted by $0 < \mathcal{K}_1^i < \dots < \mathcal{K}_j^i < \dots < \mathcal{K}_{n_i}^i$ and the corresponding bid (resp. ask) prices are denoted by $(\mathcal{C}_j^{i,BID})_{1 \leq j \leq n_i}$ (resp. $(\mathcal{C}_j^{i,ASK})_{1 \leq j \leq n_i}$).

We also assume that the stock continuously pays a deterministic dividend q_t which is automatically reinvested in shares and that the short interest rate is a deterministic function of time r_t . The corresponding factors will be denoted as follows:

$$\begin{aligned} D_r(t_1, t_2) &\equiv e^{-\int_{t_1}^{t_2} r_u du} & D_r^i &\equiv D_r(0, T_i) \\ D_{-q}(t_1, t_2) &\equiv e^{\int_{t_1}^{t_2} q_u du} & D_{-q}^i &\equiv D_{-q}(0, T_i) \\ D_{r-q}(t_1, t_2) &\equiv e^{-\int_{t_1}^{t_2} (r_u - q_u) du} & D_{r-q}^i &\equiv D_{r-q}(0, T_i) . \end{aligned}$$

By convention, we add a call struck at $\mathcal{K}_0^i \equiv 0$ for each maturity T_i : $\mathcal{C}_0^{i,BID} \equiv S_0/D_{-q}^i$ and $\mathcal{C}_0^{i,ASK} \equiv S_0/D_{-q}^i$, S_0 being the initial stock price.

Finally, we introduce some notations which will allow us to work in the forward measure:

$$K_j^i \equiv D_{r-q}^i \mathcal{K}_j^i \quad \mathcal{C}_j^{i,BID} \equiv D_{-q}^i \mathcal{C}_j^{i,BID} \quad \mathcal{C}_j^{i,ASK} \equiv D_{-q}^i \mathcal{C}_j^{i,ASK} . \quad (1)$$

2.2 Intuition and definitions

Before introducing some definitions which generalize the notions of vertical, butterfly and calendar spreads, let us try to give some intuition on the conditions call prices should satisfy, were they calibrated by an arbitrage-free model.

For simplicity, we assume in the following three examples that interest rates and dividends are zero and that the bid and ask call prices are identical. In such a case, we know that the stock is a martingale under the risk-neutral measure – see Harrison and Kreps (1979) and Harrison and Pliska (1981). Moreover if $(S_t)_{t \geq 0}$ is a martingale, then the corresponding call price function $C(K, T) \equiv \mathbb{E}[(S_T - K)^+]$ is non-increasing and convex in strike, as well as non-decreasing in maturity since $Y_t \equiv (S_t - K)^+$ is a submartingale. Consequently, if this martingale were consistent with given call quotes, these quotes would have to satisfy some constraints. For example, we can easily conclude that the call quotes of Table 1 are not compatible with the existence of a calibrated martingale. Indeed if it existed, we would have $x \geq 3$ and $x \leq 2$, what is impossible.

Maturity \ Strike	100	120
1 month	x	3
3 months	2	—

Table 1: A first example of arbitrage opportunity.

Another more complicated example is given by Table 2.

Maturity \ Strike	80	100	120
1 month	—	12	—
3 months	x_1	x_2	1
6 months	21	—	—

Table 2: A second example of arbitrage opportunity.

If there was a martingale consistent with these call quotes, then we would have:

$$\begin{aligned} 21 &\geq x_1 && \text{(The price is non-decreasing in maturity.)} \\ \frac{x_1 + 1}{2} &\geq x_2 && \text{(The price is convex in strike.)} \\ x_2 &\geq 12 && \text{(The price is non-decreasing in maturity.)} \end{aligned}$$

and this would result in $11 = \frac{21+1}{2} \geq 12$.

These two situations illustrate why the classical definitions of vertical, butterfly and calendar spreads need to be generalized when the quoted options can have any maturities and strikes. The following calendar vertical (resp. butterfly) spreads must be thought of as weighted averages of calendar spreads and classical vertical (resp. butterfly) spreads.

Definition 1 (Calendar Vertical Spreads) $\forall i_1, i_2 \in \llbracket 1, m \rrbracket$ s.t. $i_1 \leq i_2$, $\forall j_1 \in \llbracket 0, n_{i_1} \rrbracket, \forall j_2 \in \llbracket 0, n_{i_2} \rrbracket$ s.t. $K_{j_1}^{i_1} \geq K_{j_2}^{i_2}$

$$CVS_{j_1, j_2}^{i_1, i_2} \equiv C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID}$$

using Equation (1).

Definition 2 (Calendar Butterfly Spreads) $\forall i, i_1, i_2 \in \llbracket 1, m \rrbracket$ s.t. $i \leq i_1$ and $i \leq i_2$, $\forall j \in \llbracket 0, n_i \rrbracket, \forall j_1 \in \llbracket 0, n_{i_1} \rrbracket, \forall j_2 \in \llbracket 0, n_{i_2} \rrbracket$ s.t. $K_{j_1}^{i_1} < K_j^i < K_{j_2}^{i_2}$

$$CBS_{j, j_1, j_2}^{i, i_1, i_2} \equiv \frac{C_{j_1}^{i_1, ASK} - C_j^{i, BID}}{K_j^i - K_{j_1}^{i_1}} - \frac{C_j^{i, BID} - C_{j_2}^{i_2, ASK}}{K_{j_2}^{i_2} - K_j^i}$$

using Equation (1).

Note that the above definitions generalize those of classical butterfly, calendar and vertical spreads. Indeed Definition 1 is the one of a calendar spread when $i_1 < i_2$ and $K_{j_1}^{i_1} = K_{j_2}^{i_2}$. It is also the one of a vertical spread when $i_1 = i_2$. Likewise, Definition 2 is the one of a butterfly spread when $i = i_1 = i_2$.

Finally the example of Table 3 shows why the strike corresponding to the nearest maturity is always between the other two strikes in the definition of the calendar butterfly spread. Indeed, in this case, no incompatibility can be argued.

Maturity \ Strike	80	100	120
1 month	21	—	—
3 months	—	—	1
6 months	—	12	—

Table 3: Example of arbitrage-free option quotes.

3 Necessary conditions for the absence of arbitrage

In this section, we describe necessary conditions for the absence of arbitrage such as the non-negativity of calendar vertical spreads and calendar butterfly spreads, which were introduced in the previous section. We also discuss briefly why these conditions are not theoretically sufficient for the absence of arbitrage.

3.1 Assumptions

Let us assume that:

- At any time, one can take short or long positions in the stock (resp. default free bond), at the current stock (resp. bond) price, without any transaction cost or limit in the quantity;
- One can take long (resp. short) positions in the call options, at the current ask (resp. bid) price, without any transaction cost or limit in the quantity;
- Finally, the options are cash settled: one, who holds a call, receives in cash the difference between the stock price and the strike, at maturity, if this difference is positive. (Note that it is always possible to liquidate positions. Therefore this assumption does not really affect the feasibility of the strategies described in Appendix A to prove Proposition 3).

3.2 Semi-static arbitrage strategies

In the next subsection, we will show that some constraints on market quotes are necessary for the absence of arbitrage by only resorting to strategies which are semi-static. By semi-static, we mean that one is allowed to take static positions in stock, options and bonds at the initial time. Furthermore, as explained in Carr et al. (2003), one has the possibility, at the initial time, to short the stock for a given future period of time, if the stock is greater than a given value at the beginning of this period. Consequently, if a claim with the following payoff at time T_2 ($> T_1 > 0$):

$$1_{\{S_{T_1} > K\}} (S_{T_1}/D_{r-q}(T_1, T_2) - S_{T_2})$$

was available at the initial time, the strategies would be completely static. For the sake of simplicity, we will pretend, from now on, that such a costless claim exists and will be denoted by $ShortSell(T_1, T_2, K)$. Finally, notice that these strategies are a subset of the strategies, which are allowed in the Asset Pricing Theory, developed in Harrison and Kreps (1979) and Harrison and Pliska (1981). Therefore, the absence of arbitrage implies the absence of semi-static arbitrage.

3.3 Necessary conditions for the absence of semi-static arbitrage

Proposition 3 *Under the assumptions of Sections 2.1 and 3.1 and using the notations of Equation (1), the following inequalities are necessary for the absence of semi-static arbitrage:*

- $\forall i \in \llbracket 1, m \rrbracket, \forall j \in \llbracket 0, n_i \rrbracket$

$$C_j^{i,ASK} \geq 0 \tag{2}$$

$$C_0^{i,BID} - C_j^{i,ASK} \leq K_j^i \tag{3}$$

- $\forall i_1, i_2 \in \llbracket 1, m \rrbracket$ s.t. $i_1 \leq i_2, \forall j_1 \in \llbracket 0, n_{i_1} \rrbracket, \forall j_2 \in \llbracket 0, n_{i_2} \rrbracket$

$$CVS_{j_1, j_2}^{i_1, i_2} \geq 0 \text{ if } K_{j_1}^{i_1} \geq K_{j_2}^{i_2} \tag{4}$$

$$CVS_{j_1, j_2}^{i_1, i_2} > 0 \text{ if } K_{j_1}^{i_1} > K_{j_2}^{i_2} \text{ and } C_{j_1}^{i_1, BID} > 0 \tag{5}$$

- $\forall i, i_1, i_2 \in \llbracket 1, m \rrbracket$ s.t. $i \leq i_1$ and $i \leq i_2$, $\forall j \in \llbracket 0, n_i \rrbracket, \forall j_1 \in \llbracket 0, n_{i_1} \rrbracket, \forall j_2 \in \llbracket 0, n_{i_2} \rrbracket$ s.t. $K_{j_1}^{i_1} < K_j^i < K_{j_2}^{i_2}$

$$CBS_{j,j_1,j_2}^{i,i_1,i_2} \geq 0. \quad (6)$$

Proof. See Appendix A. ■

Remark 4 *The conditions of Proposition 3 are necessary for the absence of semi-static arbitrage, whatever the statistical measure of reference. But, can we claim that these conditions are sufficient? If we do not specify a statistical measure, the answer is no. Indeed, if the ask price of a call is zero and the probability that the underlying asset will be greater than the strike at maturity is positive, then to buy this call is an arbitrage strategy. Likewise, if the bid price of a call is positive and the probability that the stock will be greater than the strike at maturity is zero, then to sell this call is an arbitrage strategy.*

4 Sufficient conditions for exact calibration by an arbitrage-free model

In the previous Remark 4, we explained why the conditions of Proposition 3, which are necessary for the absence of semi-static arbitrage, are not sufficient if no assumptions about the null sets of the statistical measure are made. Since this measure is unknown a priori, it makes more sense to ask oneself if these conditions, which are obviously necessary for the existence of a consistent arbitrage-free model, are also sufficient for the existence of such a model. Proposition 5 answers this question.

Proposition 5 *Let us assume that the inequalities (2), (3), (4), (5) and (6) of Proposition 3 are satisfied. Then there exists a non-negative process $(S_t)_{t \geq 0}$, starting at S_0 , such that $(S_t e^{-\int_0^t (r_u - q_u) du})_{t \geq 0}$ is a Markov martingale and further $\forall i$ s.t. $1 \leq i \leq m$, $\forall j$ s.t. $0 \leq j \leq n_i$,*

$$\mathcal{C}_j^{i,BID} \leq \mathbb{E}[e^{-\int_0^{T_i} r_u du} (S_{T_i} - \mathcal{K}_j^i)^+] \leq \mathcal{C}_j^{i,ASK}.$$

Proof. See Appendix B. ■

5 Exact calibration to given market quotes and default probabilities

One can infer from Credit Default Swap quotes a default time distribution by using, for instance, an intensity model where default is the first jump-time of a time-inhomogeneous Poisson process. (For more details, see for example Brigo and Alfonsi (2005)). Actually, this information can be used in an equity model. Indeed, owners of a bond which is in default can make claims against the assets of the issuer to recover their loss. That is why the stock price tends to go to zero.

Therefore, if we denote the default probabilities by $P^i \equiv \mathbb{P}(\tau \leq T_i)$, where τ is the default time, and if we assume that $\tau \leq T_i \Leftrightarrow S_{T_i} = 0$, a desirable feature for an equity model could be:

$$\mathbb{P}[S_{T_i} = 0] = P^i.$$

As one might expect, not all sets of default probabilities are admissible for an arbitrage-free model, which is calibrated to given market quotes. For example, by absence of arbitrage, if the

price of a stock reaches zero, it must stay at zero. Consequently the default probabilities must be non-decreasing with maturity. We also suspect that a given default probability imposes lower bounds on put prices of the same maturity and therefore, by put-call parity, lower bounds on call prices. Proposition 6 makes these statements more precise by describing necessary and sufficient conditions on default probabilities for the existence of a calibrated arbitrage-free model.

Proposition 6 *If the conditions of Proposition 3 are satisfied, $(P^i)_{1 \leq i \leq m}$ are given real numbers, with $C_0^i \equiv C_0^{i,BID} = C_0^{i,ASK}$, it is necessary and sufficient that:*

- $$0 \leq P^1 \leq \dots \leq P^i \leq \dots \leq P^m < 1 ; \quad (7)$$

- For $i \leq k \leq m, 1 \leq j \leq n_k$,

$$P^i \leq 1 - \frac{C_0^k - C_j^{k,ASK}}{K_j^k} ; \quad (8)$$

- If $\exists i \in \llbracket 1, m \rrbracket, j \in \llbracket 1, n_i \rrbracket, k \in \llbracket i, m \rrbracket, l \in \llbracket 1, n_k \rrbracket$ such that $K_l^k > K_j^i$ and $\frac{C_0^i - C_j^{i,BID}}{K_j^i} = \frac{C_0^k - C_l^{k,ASK}}{K_l^k}$, then:

$$P^i = 1 - \frac{C_0^i - C_j^{i,BID}}{K_j^i} \quad (9)$$

for the existence of a non-negative process $(S_t)_{t \geq 0}$, starting at $S_0 > 0$, such that $(S_t e^{-\int_0^t (r_u - q_u) du})_{t \geq 0}$ is a Markov martingale and which satisfies:

$$C_j^{i,BID} \leq \mathbb{E}[e^{-\int_0^{T_i} r_u du} (S_{T_i} - K_j^i)^+] \leq C_j^{i,ASK} \text{ and } \mathbb{P}[S_{T_i} = 0] = P^i$$

for $1 \leq i \leq m, 0 \leq j \leq n_i$.

Proof. See Appendix C. ■

6 Conclusion

Using Proposition 5, we can now decide if market quotes can or cannot be calibrated by an arbitrage-free model. In the first case, semi-static arbitrage strategies may remain but their eventuality depends on the null sets of the statistical measure, which are never known in advance (see Remark 4). In the second case, we know that there exists an arbitrage opportunity which never involves more than three options (see Appendix A for a description of the corresponding semi-static arbitrage strategies).

We can also compare the equity and credit markets using Proposition 6. Indeed this proposition allows us to see if default probabilities inferred from the credit market are compatible with option prices in the framework of an arbitrage-free model. We suspect that some arbitrage strategies may hide behind the conditions of Proposition 6, if enough products are available in the credit market².

²such as Credit Default Swaps expiring at option maturities.

Finally, note that the models that we have constructed in this paper are particularly unrealistic. For instance, the number of possible states for the stock may be decreasing with maturity and the transition probability function remains largely undetermined. Buehler (2004), Laurent and Leisen (1998) and Cousot (2005) explain how to choose more realistic calibrated and arbitrage-free Markov chain models. The first working paper proposes to minimize the variance of price changes to choose the martingale transition matrices. The second proposes to choose the transition matrices by minimizing the entropy relative to a prior model. The last one proposes an algorithm also based on relative entropy minimization to add some states for the stock at each maturity, obtain smoother marginal distributions, and find transition matrices which take into account preferences on the forward volatility surface. Numerical examples illustrating the pricing of derivative securities in such a model can be found in Carr and Cousot (2005), which explains in particular how to semi-statically hedge and price a class of mildly path-dependent options.

Appendix A Proof of Proposition 3

Equations (2) and (3) as well as the non-negativity of classical vertical and butterfly spreads have already been proven in Merton (1973). Therefore let us focus on the calendar vertical and the calendar butterfly spreads.

Vertical Calendar Spreads: We show that Equations (4) and (5) are satisfied by considering the following strategy:

- Long one call, struck at $\mathcal{K}_{j_2}^{i_2}$ and expiring at T_{i_2} ;
- Short $1/D_{-q}(T_{i_1}, T_{i_2})$ call, struck at $\mathcal{K}_{j_1}^{i_1}$ and expiring at T_{i_1} ;
- If $i_1 \neq i_2$, long one unit of $ShortSell(T_{i_1}, T_{i_2}, \mathcal{K}_{j_1}^{i_1})$.

The cost of this strategy is:

$$\mathcal{C} = 1 \times \mathcal{C}_{j_2}^{i_2, ASK} - 1/D_{-q}(T_{i_1}, T_{i_2}) \times \mathcal{C}_{j_1}^{i_1, BID} + 1_{\{i_1 \neq i_2\}} \times 0 .$$

At T_{i_2} , the payoff of this strategy can be written as follows:

$$\mathcal{P} = (S_{T_{i_2}} - \mathcal{K}_{j_2}^{i_2})^+ - 1/D_{r-q}(T_{i_1}, T_{i_2}) \times (S_{T_{i_1}} - \mathcal{K}_{j_1}^{i_1})^+ + 1_{\{S_{T_{i_1}} > \mathcal{K}_{j_1}^{i_1}\}} \left(S_{T_{i_1}}/D_{r-q}(T_{i_1}, T_{i_2}) - S_{T_{i_2}} \right) .$$

In the above expression, we did not make a distinction between the cases $i_1 = i_2$ and $i_1 < i_2$, since the short-selling term vanishes if $i_1 = i_2$. We now show that this payoff is always non-negative:

$$\text{If } S_{T_{i_1}} \leq \mathcal{K}_{j_1}^{i_1},$$

$$\mathcal{P} = (S_{T_{i_2}} - \mathcal{K}_{j_2}^{i_2})^+ \geq 0$$

else,

$$\begin{aligned} \mathcal{P} &= (S_{T_{i_2}} - \mathcal{K}_{j_2}^{i_2})^+ + \mathcal{K}_{j_1}^{i_1}/D_{r-q}(T_{i_1}, T_{i_2}) - S_{T_{i_2}} \\ &\geq S_{T_{i_2}} - \mathcal{K}_{j_2}^{i_2} + \mathcal{K}_{j_1}^{i_1}/D_{r-q}(T_{i_1}, T_{i_2}) - S_{T_{i_2}} \\ &= \mathcal{K}_{j_1}^{i_1}/D_{r-q}(T_{i_1}, T_{i_2}) - \mathcal{K}_{j_2}^{i_2} \\ &= \frac{1}{D_{r-q}^{i_2}}(K_{j_1}^{i_1} - K_{j_2}^{i_2}) \\ &\geq 0 . \end{aligned}$$

We observe that this payoff is always non-negative and, moreover, it is positive with positive probability if $K_{j_1}^{i_1} > K_{j_2}^{i_2}$ and $\mathbb{P}(S_{T_{i_1}} > \mathcal{K}_{j_1}^{i_1}) > 0$. Finally, note that, under the assumption of the absence of semi-static arbitrage, $\mathbb{P}(S_{T_{i_1}} > \mathcal{K}_{j_1}^{i_1}) > 0$ is implied by $C_{j_1}^{i_1, BID} > 0$.

Calendar Butterfly Spreads: We show that Equation (6) is satisfied when $i_1 \geq i_2$ — the other case being similar. We will use the following notation:

$$N_1 \equiv \frac{D_{-q}^{i_1}}{K_j^i - K_{j_1}^{i_1}} \geq 0 \quad N_2 \equiv \frac{D_{-q}^{i_2}}{K_{j_2}^{i_2} - K_j^i} \geq 0 \quad N_i \equiv 1/D_r^i \geq 0 .$$

Consider the following strategy:

- Long N_1 calls, struck at $\mathcal{K}_{j_1}^{i_1}$, of maturity T_{i_1} ;
- Long N_2 calls, struck at $\mathcal{K}_{j_2}^{i_2}$, of maturity T_{i_2} ;
- Short $D_{-q}^i \times \left(\frac{1}{K_j^i - K_{j_1}^{i_1}} + \frac{1}{K_{j_2}^{i_2} - K_j^i} \right)$ calls, struck at $\mathcal{K}_j^i$, of maturity T_i ;
- If $i_1 > i$, long N_1 units of $ShortSell(T_i, T_{i_1}, K_{j_1}^{i_1}/D_{r-q}^i)$;
- If $i_2 > i$, long N_2 units of $ShortSell(T_i, T_{i_2}, K_{j_2}^{i_2}/D_{r-q}^i)$.

The cost of this strategy is:

$$\mathcal{C} = N_1 \times \mathcal{C}_{j_1}^{i_1, ASK} + N_2 \times \mathcal{C}_{j_2}^{i_2, ASK} - D_{-q}^i \times \left(\frac{1}{K_j^i - K_{j_1}^{i_1}} + \frac{1}{K_{j_2}^{i_2} - K_j^i} \right) \times \mathcal{C}_j^{i, BID} .$$

The payoff of this strategy, at time T_{i_1} , can be written as a weighted average of payoffs corresponding to the following strategies:

- The strategy corresponding to the calendar spread involving the call options $(T_i, K_{j_1}^{i_1}/D_{r-q}^i)$ and $(T_{i_1}, \mathcal{K}_{j_1}^{i_1})$, whose payoff at time T_{i_1} is:

$$\begin{aligned} \mathcal{P}_1 &= (S_{T_{i_1}} - \mathcal{K}_{j_1}^{i_1})^+ - 1/D_{r-q}(T_i, T_{i_1}) \left(S_{T_i} - K_{j_1}^{i_1}/D_{r-q}^i \right)^+ \\ &\quad + 1_{\{S_{T_i} > K_{j_1}^{i_1}/D_{r-q}^i\}} \left(S_{T_i}/D_{r-q}(T_i, T_{i_1}) - S_{T_{i_1}} \right) . \end{aligned}$$

- The strategy corresponding to the calendar spread involving the call options $(T_i, K_{j_2}^{i_2}/D_{r-q}^i)$ and $(T_{i_2}, \mathcal{K}_{j_2}^{i_2})$, whose payoff at time T_{i_2} is:

$$\begin{aligned} \mathcal{P}_2 &= (S_{T_{i_2}} - \mathcal{K}_{j_2}^{i_2})^+ - 1/D_{r-q}(T_i, T_{i_2}) \left(S_{T_i} - K_{j_2}^{i_2}/D_{r-q}^i \right)^+ \\ &\quad + 1_{\{S_{T_i} > K_{j_2}^{i_2}/D_{r-q}^i\}} \left(S_{T_i}/D_{r-q}(T_i, T_{i_2}) - S_{T_{i_2}} \right) . \end{aligned}$$

- The strategy corresponding to the butterfly spread, at maturity T_i , involving the strikes $K_{j_1}^{i_1}/D_{r-q}^i$, $\mathcal{K}_j^i$ and $K_{j_2}^{i_2}/D_{r-q}^i$, whose payoff at time T_i is:

$$\mathcal{P}_3 = \frac{(S_{T_i} - K_{j_1}^{i_1}/D_{r-q}^i)^+ - (S_{T_i} - \mathcal{K}_j^i)^+}{\mathcal{K}_j^i - K_{j_1}^{i_1}/D_{r-q}^i} - \frac{(S_{T_i} - \mathcal{K}_j^i)^+ - (S_{T_i} - K_{j_2}^{i_2}/D_{r-q}^i)^+}{K_{j_2}^{i_2}/D_{r-q}^i - \mathcal{K}_j^i} .$$

Indeed, the payoff of the above defined strategy, at time T_{i_1} , can be written as:

$$\mathcal{P} = N_1 \times \mathcal{P}_1 + N_2/D_r(T_{i_2}, T_{i_1}) \times \mathcal{P}_2 + N_i/D_r(T_i, T_{i_1}) \times \mathcal{P}_3 .$$

We know that $\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3 \geq 0$, whatever the values of S_{T_i} , $S_{T_{i_1}}$ or $S_{T_{i_2}}$. Consequently, $\mathcal{P} \geq 0$. Under the assumption of absence of semi-static arbitrage, this implies that $\mathcal{C} \geq 0$, which is equivalent to Equation (6).

Appendix B Proof of Proposition 5

Step 1: Change to the forward measure. If we prove under the assumptions of Proposition 5 that there exists a non-negative Markov martingale $(M_t)_{t \geq 0}$, starting at S_0 , which satisfies:

$$C_j^{i,BID} \leq \mathbb{E}[(M_{T_i} - K_j^i)^+] \leq C_j^{i,ASK} \quad (10)$$

for all $1 \leq i \leq m$, $0 \leq j \leq n_i$ then the conclusion follows since Equation (10) can be rewritten:

$$C_j^{i,BID} \leq D_r^i \mathbb{E}[(M_{T_i}/D_{r-q}^i - K_j^i)^+] \leq C_j^{i,ASK}$$

using the definitions of Equation (1). Consequently, the process

$$(S_t) \equiv (M_t/D_{r-q}(0, t))$$

is the perfect candidate.

Another way to formulate it is to say that we have to prove Proposition 5 in the case where interest rates, dividends are zero and the market quotes $(C_j^{i,BID})_{1 \leq i \leq m, 1 \leq j \leq n_i}$, $(C_j^{i,ASK})_{1 \leq i \leq m, 1 \leq j \leq n_i}$, and strikes $(K_j^i)_{1 \leq i \leq m, 1 \leq j \leq n_i}$ are replaced respectively by $(C_j^{i,BID})_{1 \leq i \leq m, 1 \leq j \leq n_i}$, $(C_j^{i,ASK})_{1 \leq i \leq m, 1 \leq j \leq n_i}$ and $(K_j^i)_{1 \leq i \leq m, 1 \leq j \leq n_i}$.

Step 2: Kellerer's theorem. Now, if we are able to construct marginal distributions which:

- are consistent with the new call quotes $(C_j^{i,ASK})_{1 \leq i \leq m, 1 \leq j \leq n_i}$ and $(C_j^{i,BID})_{1 \leq i \leq m, 1 \leq j \leq n_i}$,
- are non-decreasing in the convex order (see Theorem 7 for the exact mathematical formulation),
- have the same mean S_0 ,

then we can use Kellerer's theorem to conclude that there exists a martingale consistent with all these marginal distributions and therefore with the call quotes.

Theorem 7 (Kellerer's theorem (1972)) *Let $(\mu_t)_{t \in [0, T]}$ be a family of probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with first moment, such that, for $s < t$, μ_t dominates μ_s in the convex order, i.e. for each convex function $\phi: \mathbb{R} \rightarrow \mathbb{R}$, μ_t -integrable for each $t \in [0, T]$, we have:*

$$\int_{\mathbb{R}} \phi d\mu_t \geq \int_{\mathbb{R}} \phi d\mu_s .$$

Then there exists a Markov process $(M_t)_{t \in [0, T]}$ with these marginal distributions under which it is a submartingale. Furthermore if the means are independent of t then $(M_t)_{t \in [0, T]}$ is a martingale.

Proof. See Kellerer (1972), p. 120. ■

Therefore we are left with constructing consistent marginal distributions which are non-decreasing in the convex order (henceforth NDCO) and have the same mean. To do so, we will actually construct non-decreasing risk-neutral call price functions and use Lemma 8 which explains how to associate in a one-to-one way discrete distributions to a particular class of call price functions.

Step 3: One-to-one relationship between certain types of call price functions and discrete distributions. The following lemma is simply an application of the well-known results of Breeden and Litzenberger (1978).

Lemma 8 (Discrete Distribution) *If $N \in \mathbb{N}^*$, $k_0 = 0 < \dots < k_j < \dots < k_N$ are $N + 1$ real numbers and $C : \mathbb{R}_+ \rightarrow \mathbb{R}$ is a function, which is continuous, convex, linear and decreasing on each interval $[k_j; k_{j+1}]$ with $0 \leq j \leq N - 1$ (with a slope greater than or equal to -1 on $[k_0; k_1]$) and zero after k_N then the following distribution*

$$\mu = \sum_{j=0}^N q_j \delta_{k_j} \quad \text{with} \quad q_0 = 1 - \frac{P(k_0) - P(k_1)}{k_1 - k_0},$$

$$q_j = \frac{P(k_{j-1}) - P(k_j)}{k_j - k_{j-1}} - \frac{P(k_j) - P(k_{j+1})}{k_{j+1} - k_j} \quad \text{for } 1 \leq j \leq N - 1 \quad \text{and} \quad q_N = \frac{P(k_{N-1})}{k_N - k_{N-1}}$$

is such that:

$$C(K) = \int_0^\infty (x - K)^+ d\mu \quad \text{for all } K \in \mathbb{R}_+. \quad (11)$$

In the next step, we will use Lemma 8 to construct risk-neutral marginal distributions. At a given maturity, the quoted strikes will be among $\{k_j\}_{0 \leq j \leq N}$ and the corresponding prices will be given by the function C .

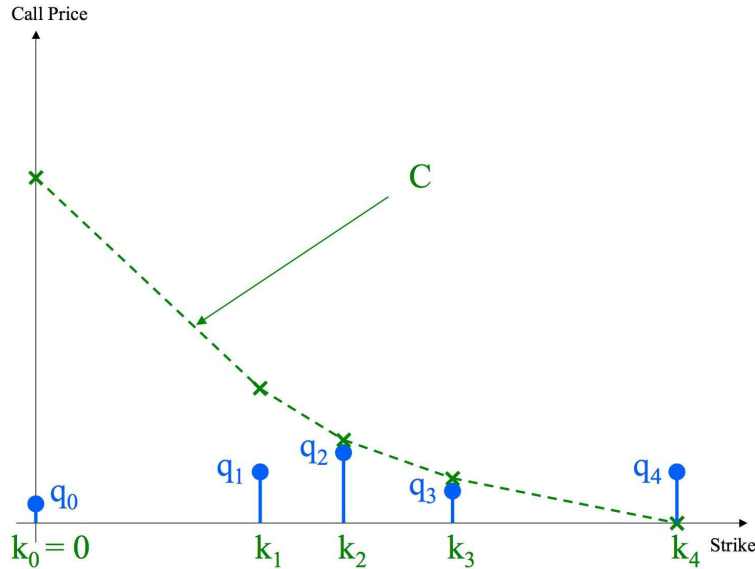


Figure 1: Sketch of a possible discrete distribution.

Step 4: Construction of consistent call price functions, with the right properties. In the special case where bid and ask prices are identical, a first guess for a consistent call price function at maturity T_i could be a continuous piecewise linear function, which connects the prices at this maturity. However this construction does not result in call price functions which are non-decreasing with maturity, as can be observed in the simple example of Table 4, where $S_0 = 100$.

Maturity \ Strike	80	100
1 month	—	30
3 months	35	—

Table 4: A situation where option quotes with longer maturities matter.

That is why our method for building a call price function at maturity T_i will take into account call options which expire at a later maturity. (See Equation (16)).

Another important issue is to specify the point where the call price function at maturity T_i will reach zero. This point must be large enough to preserve the convexity of the call price function but small enough to allow the call price function to be below the ask quotes at maturity greater than or equal to T_i .

Consider:

$$Slopes1 \equiv \left\{ \frac{C_{j_1}^{i_1, ASK} - C_{j_2}^{i_2, ASK}}{K_{j_1}^{i_1} - K_{j_2}^{i_2}} < 0 \mid 1 \leq i_1 \leq i_2 \leq m, 0 \leq j_1 \leq n_{i_1}, 0 \leq j_2 \leq n_{i_2} \text{ s.t. } K_{j_1}^{i_1} > K_{j_2}^{i_2} \right\} \quad (12)$$

$$Slopes2 \equiv \left\{ \frac{C_{j_1}^{i_1, BID} - C_{j_2}^{i_2, ASK}}{K_{j_1}^{i_1} - K_{j_2}^{i_2}} < 0 \mid 1 \leq i_1 \leq i_2 \leq m, 0 \leq j_1 \leq n_{i_1}, 0 \leq j_2 \leq n_{i_2} \text{ s.t. } K_{j_1}^{i_1} > K_{j_2}^{i_2} \right\} \quad (13)$$

and

$$MaxSlope \equiv \max \left(Slopes1 \cup Slopes2 \cup \{-1\} \right) .$$

If $C_{n_i}^{i, ASK} = 0$, then:

$$K_{n_i+1}^i \equiv K_{n_i}^i + \epsilon_i \text{ with } \epsilon_i > 0 \quad (14)$$

else:

$$K_{n_i+1}^i \equiv \min \left(\left\{ K_{j_1}^{i_1} - C_{j_1}^{i_1, ASK} / (\alpha_i MaxSlope) \mid i \leq i_1, 0 \leq j_1 \leq n_{i_1}, C_{j_1}^{i_1, ASK} > 0 \right\} \right) \quad (15)$$

with $\alpha_i \in (0, 1)$.

$$\forall i \in \llbracket 1, m \rrbracket, \forall j \in \llbracket 0, n_i \rrbracket,$$

$$\begin{aligned} A_j^{i, BID} &\equiv (K_j^i, C_j^{i, BID}) & A_j^{i, ASK} &\equiv (K_j^i, C_j^{i, ASK}) \\ A_{n_i+1}^{i, BID} &\equiv (K_{n_i+1}^i, 0) & A_{n_i+1}^{i, ASK} &\equiv (K_{n_i+1}^i, 0) . \end{aligned}$$

Let us define $C^i(K)$ for $K \in \mathbb{R}_+$ as

$$\min \left\{ y \mid \begin{array}{l} \forall (a, b) \in \mathbb{R}_- \times \mathbb{R} \text{ s.t.} \\ \left(\forall i_1 \in \llbracket i, m \rrbracket, \forall j_1 \in \llbracket 0, n_{i_1} + 1 \rrbracket, aK_{j_1}^{i_1} + b \leq C_{j_1}^{i_1, ASK} \right) \\ \Rightarrow aK + b \leq y \end{array} \right\} \quad (16)$$

and $\forall i \in \llbracket 1, m \rrbracket, \forall j \in \llbracket 0, n_i + 1 \rrbracket$,

$$C_j^i \equiv C^i(K_j^i) \quad A_j^i \equiv (K_j^i, C_j^i) . \quad (17)$$

Informally, C^i is the low boundary of the convex hull of the set of points:

$$B^i = \{A_{j_1}^{i_1, ASK} \text{ s.t. } i \leq i_1 \leq m, 0 \leq j_1 \leq n_{i_1} + 1\} . \quad (18)$$

Note that a cautious choice of $K_{n_i+1}^i$ is important if $C_{n_i}^{i, ASK} > 0$. First, the ask prices do not need

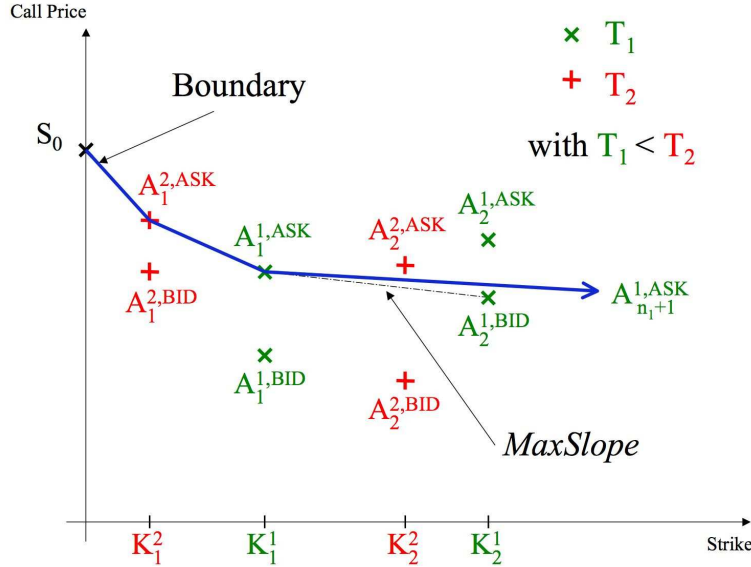


Figure 2: Boundary at T_1 .

to be non-increasing, as it can be observed in Figure 2. That is why $K_{n_i+1}^i$ is defined as a minimum in Equation (15). Second, $MaxSlope$ must be large enough to ensure the convexity of the boundary. This explains why it must be greater than or equal to the elements of $Slopes1$ (see Equation (12)). Finally, $MaxSlope$ needs to be large enough to have the call price function greater than the bid prices. That is why it must be at least equal to the maximum of $Slopes2$ (see Equation (13)).

Notice, that in the next six steps, $[P_1; P_2]$ will denote the segment linking the points P_1 and P_2 which belong to $(\mathbb{R}_+)^2$.

Step 4.1: A first sanity check. First, we show that $K_{n_{i_1}+1}^{i_1} > K_j^i$ if $i_1 \geq i, 0 \leq j \leq n_i$ and $C_j^{i, BID} > 0$. Let us assume that $K_{n_{i_1}+1}^{i_1} \leq K_j^i$. We have the following two cases:

- If $C_{n_{i_1}}^{i_1, ASK} = 0$, the condition on the calendar (vertical) spread between the points $A_{n_{i_1}}^{i_1, ASK}$ and $A_j^{i, BID}$ imposes that $C_j^{i, BID} \leq 0$, which is a contradiction.
- If $C_{n_{i_1}}^{i_1, ASK} > 0$, let us denote by $A_{j^*}^{i^*}$ a point such that $i_1 \leq i^*, j^* \leq n_{i^*}$ and such that the slope of the segment $[A_{j^*}^{i^*, ASK}; A_{n_{i_1}+1}^{i_1, ASK}]$ is equal to $\alpha_{i_1} MaxSlope$. The slope of $[A_{j^*}^{i^*, ASK}; A_j^{i, BID}]$ is negative since $C_j^{i, BID} > 0$ and is less than $MaxSlope$ by definition. But this slope is also greater than $\alpha_{i_1} MaxSlope$ since the segment $[A_{j^*}^{i^*, ASK}; A_j^{i, BID}]$ is above the segment $[A_{j^*}^{i^*, ASK}; A_{n_{i_1}+1}^{i_1, ASK}]$, which is a contradiction.

Consequently, $K_{n_{i_1}+1}^{i_1} > K_j^i$.

Step 4.2: Immediate properties of the boundary. It is clear that C^i is *continuous, piecewise linear, convex*, and that its nodes are among the $K_{j_1}^{i_1}$ (with corresponding value $C_{j_1}^{i_1, ASK}$), $i \leq i_1 \leq m, 0 \leq j_1 \leq n_{i_1} + 1$.

Step 4.3: We show that the slope of the boundary on the first segment is greater than or equal to -1. The value of the boundary in $K_0^i = 0$ is clearly $S_0 = C_0^{i, ASK} = C_0^{i, BID}$. Let us denote by $K_{j_1}^{i_1}$ the smallest non-zero node of the boundary. We have the following cases:

- If $j_1 \leq n_{i_1}$, then the slope is greater than or equal to -1 because the vertical spread involving the calls expiring at T_{i_1} and struck at $K_0^{i_1} = 0$ and $K_{j_1}^{i_1}$ is bounded.

$$\frac{C_{j_1}^{i_1, ASK} - C_0^{i_1, BID}}{K_{j_1}^{i_1}} = \frac{C_{j_1}^{i_1, ASK} - C_0^{i_1, BID}}{K_{j_1}^{i_1}} \geq -1.$$

- If $j_1 = n_{i_1} + 1$, then the slope is equal to $\alpha_{i_1} \text{MaxSlope}$ and is bounded by $-\alpha_{i_1} > -1$ by definition of *MaxSlope*.

Step 4.4: We show that the boundary is zero after a given value. Since $C^i(K_{n_i+1}^i) = 0$ and the linear functions involved in the definition of C^i (Equation (16)) have non-positive slopes, the boundary is zero on $[K_{n_i+1}^i, +\infty)$.

Step 4.5: We study the monotonicity of C^i . We consider two consecutive nodes of the boundary $K_{j_1}^{i_1}$ and $K_{j_2}^{i_2}$ with $K_{j_1}^{i_1} < K_{j_2}^{i_2}$. Denote by $K_{j^*}^{i^*}$ the smallest strike which corresponds to a node of the boundary whose call value is zero, with $i \leq i^*$ and $0 \leq j^* \leq n_{i^*} + 1$. Since we know that the boundary is flat at the right of $K_{j^*}^{i^*}$, we only need to prove that the segment $[A_{j_1}^{i_1, ASK}; A_{j_2}^{i_2, ASK}]$ is decreasing if $K_{j_2}^{i_2} \leq K_{j^*}^{i^*}$.

We know that $[A_{j_1}^{i_1, ASK}; A_{j_2}^{i_2, ASK}]$ is below $[A_{j_1}^{i_1, ASK}; A_{j^*}^{i^*, ASK}]$ since $A_{j_2}^{i_2, ASK}$ is on the boundary. Moreover since $K_{j_1}^{i_1} < K_{j^*}^{i^*}$, $C_{j_1}^{i_1, ASK} > 0$, which implies that $[A_{j_1}^{i_1, ASK}; A_{j^*}^{i^*, ASK}]$ has a negative slope. Consequently, $[A_{j_1}^{i_1, ASK}; A_{j_2}^{i_2, ASK}]$ has a negative slope.

Step 4.6: We show that the boundary is consistent with the call quotes. Let us prove that $C_j^{i, BID} \leq C_j^i \leq C_j^{i, ASK}$ for $0 \leq j \leq n_i$. Because of the definition of C_j^i , it is clear that $C_j^i \leq C_j^{i, ASK}$. Consequently, we only need to show that $C_j^{i, BID} \leq C_j^i$. If $j = 0$ or if $C_j^{i, BID} \leq 0$, the conclusion is obvious since we have $C_0^i = C_0^{i, ASK} = C_0^{i, BID}$ in the first case, and $C_j^i \geq 0$ in the second. Therefore, let us consider a point $A_j^{i, ASK}$ ($1 \leq j \leq n_i$) such that $C_j^{i, BID} > 0$.

First, let us assume that K_j^i is a node of the boundary and denote by $C_{j_1}^{i_1, ASK}$, the corresponding value with $i_1 > i$. (The bid price is less than or equal to the ask price. Consequently we can eliminate the trivial case where $i_1 = i$). Thanks to Step 4.1, we know that $j_1 \leq n_{i_1}$. Consequently, the condition on the calendar spread ensures that $C_{j_1}^{i_1, BID} \leq C_{j_1}^{i_1, ASK}$.

If K_j^i is not a node of the boundary, denote by $K_{j_1}^{i_1}$ (resp. $K_{j_2}^{i_2}$) the greatest (resp. smallest) node of the boundary which is less (resp. greater) than K_j^i . ($K_{j_2}^{i_2}$ exists since $K_{n_i+1}^i$ is on the boundary). Recall that $i \leq i_1$, $i \leq i_2$, $j_1 \leq n_{i_1}$ and $K_{j_1}^{i_1} < K_j^i < K_{j_2}^{i_2}$.

Due to the definition of C^i , we just need to prove that $A_j^{i, BID}$ is located below or on the segment $[A_{j_1}^{i_1, ASK}; A_{j_2}^{i_2, ASK}]$.

- If $j_2 \leq n_{i_2}$, the non-negativity of the calendar butterfly spreads ensures that $A_j^{i,BID}$ is below or on the segment $[A_{j_1}^{i_1,ASK}; A_{j_2}^{i_2,ASK}]$.
- If $j_2 = n_{i_2} + 1$, then we have the following two cases:
 - If $C_{n_{i_2}}^{i_2,ASK} = 0$, then $A_{n_{i_2}}^{i_2,ASK}$ is also on the boundary. Since $A_{j_1}^{i_1,ASK}$ and $A_{n_{i_2}+1}^{i_2,ASK}$ are consecutive on the boundary, $C_{j_1}^{i_1,ASK} = 0$. The condition on the calendar vertical spreads between the points $A_{j_1}^{i_1,ASK}$ and $A_j^{i,BID}$ imposes that $C_j^i \leq 0$, which is a contradiction.
 - If $C_{n_{i_2}}^{i_2,ASK} > 0$, denote by $A_{j_2^*}^{i_2^*,ASK}$ a point such that $i \leq i_2 \leq i_2^*$, $j_2^* \leq n_{i_2^*}$ and such that the slope of the segment $[A_{j_2^*}^{i_2^*,ASK}; A_{n_{i_2}+1}^{i_2,ASK}]$ is equal to $\alpha_{i_2} \text{MaxSlope}$. The slope of the segment $[A_{j_1}^{i_1,ASK}; A_{n_{i_2}+1}^{i_2,ASK}]$ must be greater than or equal to the slope of the segment $[A_{j_2^*}^{i_2^*,ASK}; A_{n_{i_2}+1}^{i_2,ASK}]$, since $A_{j_1}^{i_1,ASK}$ and $A_{n_{i_2}}^{i_2,ASK}$ are two consecutive nodes on the boundary. Its slope is therefore greater than or equal to $\alpha_{i_2} \text{MaxSlope}$. If $A_j^{i,BID}$ is not below or on the segment $[A_{j_1}^{i_1,ASK}; A_{n_{i_2}+1}^{i_2,ASK}]$, the slope of the segment $[A_{j_1}^{i_1,ASK}; A_j^{i,BID}]$ is greater than the slope of the segment $[A_{j_1}^{i_1,ASK}; A_{n_{i_2}+1}^{i_2,ASK}]$, and consequently greater than $\alpha_{i_2} \text{MaxSlope}$. On the other hand, since $C_j^{i,BID} > 0$, the slope of $[A_{j_1}^{i_1,ASK}; A_j^{i,BID}]$ is negative because of the condition on the calendar vertical spread and must therefore be less than or equal to MaxSlope , which is a contradiction. Consequently, $A_j^{i,BID}$ is below or on the segment $[A_{j_1}^{i_1,ASK}; A_{n_{i_2}+1}^{i_2,ASK}]$.

We have proved that C^i satisfies all the conditions of Lemma 8. Therefore we can associate to it a distribution μ_i satisfying Equation (11) of Lemma 8. Moreover this distribution is risk-neutral for the calls of maturity T_i since $C_j^{i,BID} \leq C_j^i \leq C_j^{i,ASK}$.

Step 5: We prove that the distributions are NDCO and that their means are constant over time. For a given strike, the prices are non-decreasing with maturity since $B^{i+1} \subseteq B^i$ implies that $C^i \leq C^{i+1}$, $1 \leq i < m$. (See Equation (18) for the definition of B^i). Moreover, $C^0(K) = (S_0 - K)^+$ for $K \in \mathbb{R}_+$. Therefore it is less than C^1 , which is equal to S_0 in $K = 0$, has a slope in $0+$ greater than or equal to -1 , is convex and non-negative. Finally the prices of European put options are also non-decreasing with maturity because of put-call parity and since any convex function can be approximated by linear combinations with positive weights of put and call functions, the distributions are NDCO.

Finally the means of the different distributions are constant over time because all the calls struck at 0 have the same price.

Remark 9 Note that we have some freedom in the choice of $(K_{n_i+1}^i)_{1 \leq i \leq m}$. In particular, if we want them all to be equal to a given value, this is possible by choosing α_i small enough or ϵ_i big enough for each $i \in \llbracket 1, m \rrbracket$. (See Equations (14) and (15) for the definitions of $K_{n_i+1}^i$).

Remark 10 One may ask if relaxing the assumption of no bid-ask spread on the stock is possible. The proof of Proposition 5 remains valid, in the presence of a bid-ask spread for the stock, if we replace the condition $C_0^{i,BID} - C_j^{i,ASK} \leq K_j^i$, $1 \leq i \leq m$, $1 \leq j \leq n_i$, by $C_0^{i,ASK} - C_j^{i,ASK} \leq K_j^i$. Nevertheless, these conditions, as well as the cross-maturity inequalities, are not necessary for the absence of semi-static arbitrage anymore. Actually, the inequalities which are necessary for the

existence of a consistent martingale are stronger than the conditions implied by the absence of arbitrage in this case.

Appendix C Proof of Proposition 6

Step 1: Change to the forward measure. As in Step 1 of Appendix B, we work in the forward measure. If we imagine that the only available market prices are $C_j^{i,BID}$ (resp. $C_j^{i,ASK}$) for the bid (resp. ask) price of a call option of maturity T_i and struck at K_j^i for $1 \leq i \leq m$, $0 \leq j \leq n_i$, then we can assume that interest rates and dividends are zero.

Step 2: Let us show that the conditions of Proposition 6 are necessary.

Step 2.1: Condition (7). The fact that the default probabilities are non-decreasing is a direct consequence of the fact that a non-negative martingale, which is at zero stays at zero. The default probabilities are also less than 1 because we assume that $S_0 > 0$.

Step 2.2: Condition (8) is necessary because imposing the probability in zero gives a lower bound for the ask price of a call option. Indeed, for $1 \leq i \leq k \leq m$ and $1 \leq l \leq n_k$, we have:

$$\begin{aligned} C_l^{k,ASK} &\geq \mathbb{E}[(S_{T_k} - K_l^k)^+] \\ &= \mathbb{E}[(S_{T_k} - K_l^k) + (K_l^k - S_{T_k})^+] \\ &\geq S_0 - K_l^k + \mathbb{P}(S_{T_k} = 0)K_l^k \\ &= S_0 - K_l^k + P^k K_l^k \\ &\geq S_0 - K_l^k + P^i K_l^k. \end{aligned}$$

Step 2.3: Condition (9) is necessary because, in this situation, the probability for the stock to be between 0 and K_l^k is zero. Indeed the value of the butterfly spread at maturity T_i involving the strikes 0, K_j^i and K_l^k is zero.

$$\begin{aligned} &\mathbb{E} \left(\frac{S_{T_i} - (S_{T_i} - K_j^i)^+}{K_j^i} - \frac{(S_{T_i} - K_j^i)^+ - (S_{T_i} - K_l^k)^+}{K_l^k - K_j^i} \right) \\ &\leq \mathbb{E} \left(\frac{S_{T_i} - (S_{T_i} - K_j^i)^+}{K_j^i} - \frac{(S_{T_i} - K_j^i)^+ - (S_{T_k} - K_l^k)^+}{K_l^k - K_j^i} \right) \\ &\leq \frac{S_0 - C_j^{i,BID}}{K_j^i} - \frac{C_j^{i,BID} - C_l^k}{K_l^k - K_j^i} \\ &= 0 \end{aligned}$$

since $((S_t - K)^+)$ is a submartingale. As a consequence, the bid price of the call struck at K_j^i and maturity T_i has the following upper bound:

$$\begin{aligned} C_j^{i,BID} &\leq \mathbb{E}[(S_{T_i} - K_j^i)^+] \\ &= \mathbb{E}[(S_{T_i} - K_j^i) + (K_j^i - S_{T_i})^+] \\ &= S_0 - K_j^i + P^i K_j^i. \end{aligned}$$

Therefore, $P^i \geq 1 - (S_0 - C_j^{i,BID})/K_j^i$. Finally, the condition $P^i \leq 1 - (S_0 - C_l^{k,ASK})/K_l^k$ allows us to conclude that Condition (9) holds.

Step 3: The conditions of Proposition 6 are sufficient. The idea is to add fictitious quotes at a strike, that is smaller than all the strikes, to calibrate the default probabilities and conserve the features of the call price functions constructed in Appendix B. Let us denote this strike by K_* and define the bid (resp. ask) price of the call option struck at this strike and maturity T_i by:

$$C_*^i \equiv C_*^{i,ASK} \equiv C_*^{i,BID} \equiv S_0 - (1 - P^i)K_* . \quad (19)$$

We see that if K_* goes to zero, the call price functions with the added quotes tend to the previous call price functions. That is why we are confident that for K_* small enough, the call price functions with the added quotes will have the right features.

Moreover if the model is calibrated to this fictitious quote then we will have:

$$\begin{aligned} S_0 - (1 - P^i)K_* &= C_*^i \\ &= \mathbb{E}[(S_{T_i} - K_*)^+] \\ &= S_0 - K_* + \mathbb{E}[(K_* - S_{T_i})^+] \\ &= S_0 - K_* + \mathbb{P}(S_{T_i} = 0)K_* \\ &= S_0 - (1 - \mathbb{P}(S_{T_i} = 0))K_* \end{aligned}$$

which implies $\mathbb{P}(S_{T_i} = 0) = P^i$. See Figure 3 for an example.

Step 3.1: Adding fictitious quotes. Let us define:

$$K_*^0 \equiv \min \left(\left\{ \frac{S_0}{1 - P^i} \mid 1 \leq i \leq m \right\} \right) \quad (20)$$

$$K_*^1 \equiv \min \left(\left\{ \frac{S_0 - C_j^{i,BID}}{1 - P^i} \mid 1 \leq i \leq m, 1 \leq j \leq n_i \right\} \right) \quad (21)$$

$$\begin{aligned} K_*^2 \equiv \min \left(\left\{ \frac{(S_0 - C_{j_1}^{i_1,BID})K_{j_2}^{i_2} - (S_0 - C_{j_2}^{i_2,ASK})K_{j_1}^{i_1}}{C_{j_2}^{i_2,ASK} - C_{j_1}^{i_1,BID} + (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1})} \mid 1 \leq i_1 \leq i_2 \leq m, 1 \leq j_1 \leq n_{i_1}, \dots \right. \right. \\ \left. \dots 1 \leq j_2 \leq n_{i_2}, \text{ s.t. } K_{j_1}^{i_1} < K_{j_2}^{i_2} \text{ and } (S_0 - C_{j_1}^{i_1,BID})K_{j_2}^{i_2} > (S_0 - C_{j_2}^{i_2,ASK})K_{j_1}^{i_1} \right\} \right) \end{aligned} \quad (22)$$

$$K_*^3 \equiv \min \left(\{K_j^i \mid 1 \leq i \leq m, 1 \leq j \leq n_i\} \right) \quad (23)$$

with the convention $\min(\emptyset) = +\infty$.

K_*^0 and K_*^1 are well defined and positive since we ignore the trivial case where $S_0 = 0$. As far as K_*^2 is concerned, let us consider $1 \leq i_1 \leq i_2 \leq m$, $1 \leq j_1 \leq n_{i_1}$, $1 \leq j_2 \leq n_{i_2}$ such that $K_{j_1}^{i_1} < K_{j_2}^{i_2}$ and $(S_0 - C_{j_1}^{i_1,BID})K_{j_2}^{i_2} > (S_0 - C_{j_2}^{i_2,ASK})K_{j_1}^{i_1}$. We have the following cases:

- If $S_0 \leq C_{j_2}^{i_2,ASK}$, then clearly

$$\begin{aligned} &(S_0 - C_{j_1}^{i_1,BID})K_{j_2}^{i_2} - (S_0 - C_{j_2}^{i_2,ASK})K_{j_1}^{i_1} \\ &\geq (S_0 - C_{j_1}^{i_1,BID})K_{j_2}^{i_2} \\ &> 0 . \end{aligned}$$

Indeed, the last inequality is a consequence of the positivity of the vertical spread if $C_{j_1}^{i_1,BID} > 0$ and of the positivity of S_0 if $C_{j_1}^{i_1,BID} \leq 0$. As far as the denominator is concerned, we have:

$$\begin{aligned} &C_{j_2}^{i_2,ASK} - C_{j_1}^{i_1,BID} + (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1}) \\ &> (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1}) \\ &> 0 . \end{aligned}$$

- If $S_0 > C_{j_2}^{i_2, ASK}$, then we have:

$$\begin{aligned}
& C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID} + (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1}) \\
& \geq C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID} + (1 - P^{i_2})(K_{j_2}^{i_2} - K_{j_1}^{i_1}) \\
& \geq C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID} + \frac{S_0 - C_{j_2}^{i_2, ASK}}{K_{j_2}^{i_2}}(K_{j_2}^{i_2} - K_{j_1}^{i_1}) \\
& = (K_{j_2}^{i_2} - K_{j_1}^{i_1}) \left(\frac{S_0 - C_{j_2}^{i_2, ASK}}{K_{j_2}^{i_2}} - \frac{C_{j_1}^{i_1, BID} - C_{j_2}^{i_2, ASK}}{K_{j_2}^{i_2} - K_{j_1}^{i_1}} \right).
\end{aligned}$$

Moreover, the last quantity is positive if $(S_0 - C_{j_1}^{i_1, BID})K_{j_2}^{i_2} > (S_0 - C_{j_2}^{i_2, ASK})K_{j_1}^{i_1} > 0$. Indeed, if $d > b > 0$ and $a/b > c/d > 0$, then $(c - a)/(d - b) < c/d$.

Consequently, K_*^2 is well defined and positive. Let us assume that K_* is a point of the interval $(0, \min(K_*^0, K_*^1, K_*^2, K_*^3))$.

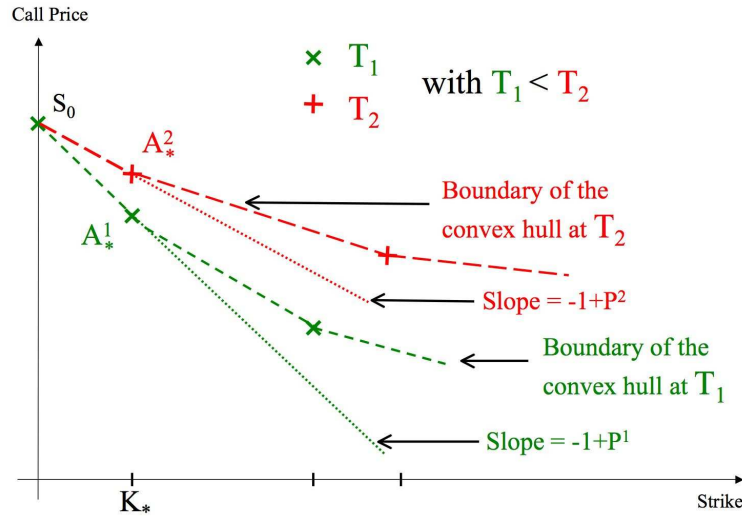


Figure 3: Fictitious quotes introduced to calibrate the default probabilities.

Step 3.2: We prove that the market quotes along with the new quotes satisfy the assumptions of Proposition 5.

Step 3.2.1: The added call quotes are non-negative. For $1 \leq i \leq m$, we have:

$$C_*^i = S_0 - (1 - P^i)K_* > S_0 - (1 - P^i)K^0 \geq 0.$$

Step 3.2.2: The slope of the call price function at the right of 0 is less than or equal to -1.

$$C_0^i - C_*^i = (1 - P^i)K_* \leq K_*$$

Step 3.2.3: The calendar vertical spreads are non-negative. The only strike which is smaller than K_* is 0. For $1 \leq i \leq i' \leq m$, we have:

$$CVS_{*,0}^{i,i'} \equiv C_0^{i'} - C_*^i = (1 - P^i)K_* > 0 .$$

Moreover,

$$CVS_{*,*}^{i,i'} \equiv C_*^{i'} - C_*^i = (P^{i'} - P^i)K_* \geq 0 .$$

For $1 \leq i \leq k$ and $1 \leq j \leq n_i$ such that $K_* < K_j^i$:

$$\begin{aligned} CVS_{j,*}^{i,k} &\equiv C_*^k - C_j^{i,BID} \\ &\geq C_*^i - C_j^{i,BID} \\ &= S_0 - (1 - P^i)K_* - C_j^{i,BID} \\ &> S_0 - (1 - P^i)K_*^1 - C_j^{i,BID} \\ &\geq 0 \end{aligned}$$

because of the definition of K_*^1 (see Equation (21)).

Step 3.2.4: The calendar butterfly spreads are non-negative. First, we consider the calendar butterfly spread involving $A_0^{i_1}$, $A_*^i \equiv (K_*^i, C_*^i)$ and $A_{j_2}^{i_2}$ with $i \leq i_1, i_2$, $1 \leq j_2 \leq n_{i_2}$.

$$\frac{C_0^i - C_*^i}{K_*} = 1 - P^i \geq \frac{S_0 - C_{j_2}^{i_2,ASK}}{K_{j_2}^{i_2}} .$$

Second, we need to consider the calendar butterfly spread involving A_*^i , $A_{j_1}^{i_1}$ and $A_{j_2}^{i_2}$ with $i_1 \leq i, i_2$ and $1 \leq j_1 \leq n_{i_1}$ such that $K_{j_1}^{i_1} < K_{j_2}^{i_2}$. Let us show that:

$$(C_{j_2}^{i_2,ASK} - C_{j_1}^{i_1,BID} + (1 - P^i)(K_{j_2}^{i_2} - K_{j_1}^{i_1}))K_* \quad (24)$$

$$\leq K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1,BID}) - K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2,ASK}) . \quad (25)$$

We have to differentiate two cases. If $K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1,BID}) > K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2,ASK})$, then we have:

$$\begin{aligned} K_* &< K_*^2 \\ &\leq \frac{K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1,BID}) - K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2,ASK})}{C_{j_2}^{i_2,ASK} - C_{j_1}^{i_1,BID} + (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1})} \\ &\leq \frac{K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1,BID}) - K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2,ASK})}{C_{j_2}^{i_2,ASK} - C_{j_1}^{i_1,BID} + (1 - P^i)(K_{j_2}^{i_2} - K_{j_1}^{i_1})} \end{aligned}$$

and Equation (24) follows.

Let us show that Equation (24) is also valid in the case where $K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1,BID}) \leq K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2,ASK})$ for any positive value of K_* . First, in such a case, $K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1,BID}) = K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2,ASK})$

because of the non-negativity of the calendar butterfly spread involving $A_0^{i_1}$, $A_{j_1}^{i_1}$ and $A_{j_2}^{i_2}$. Moreover we have:

$$P^{i_1} = 1 - \frac{S_0 - C_{j_1}^{i_1, BID}}{K_1^{i_1}} \quad (26)$$

$$= 1 - \frac{C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID}}{K_{j_1}^{i_1} - K_{j_2}^{i_2}} \quad (27)$$

because $b \neq 0$, $d \neq 0$, $d \neq b$ and $a/b = c/d$ implies that $(a - c)/(b - d) = c/d$. From (27), we deduce that:

$$\begin{aligned} & (C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID} + (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1}))K_* \\ \leq & (C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID} + (1 - P^{i_1})(K_{j_2}^{i_2} - K_{j_1}^{i_1}))K_* \\ = & 0 \\ = & K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1, BID}) - K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2, ASK}) \end{aligned}$$

and consequently (24) holds. It follows that:

$$\begin{aligned} & (C_{j_2}^{i_2, ASK} - C_{j_1}^{i_1, BID})K_* + (K_{j_2}^{i_2} - K_{j_1}^{i_1})(S_0 - C_*) \\ \leq & K_{j_2}^{i_2}(S_0 - C_{j_1}^{i_1, BID}) - K_{j_1}^{i_1}(S_0 - C_{j_2}^{i_2, ASK}) \end{aligned}$$

from which we deduce easily that

$$\frac{C_*^i - C_{j_1}^{i_1, BID}}{K_{j_1}^{i_1} - K_*} - \frac{C_{j_1}^{i_1, BID} - C_{j_2}^{i_2, ASK}}{K_{j_2}^{i_2} - K_{j_1}^{i_1}} \geq 0 .$$

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